

4. **Door hangers and incentives induced little participation in lead water pipe inspections**, with Ludovica Gazze, Heather Klemick, and Ann Wolverton. 2025. *AWWA Water Science*. (ungated here)

5. **The social costs of hydrofluorocarbons and the large climate benefits from their expedited phaseout**, with Tammy Tan and Lisa Rennels. 2024. *Nature Climate Change*. (ungated here)

6. **Factors influencing customer participation in a program to replace lead pipes for drinking water**, with Heather Klemick, Ann Wolverton, Kristin Epstein, Sandra Kuting, and Sarah Armstrong. 2024. *Journal of Environmental and Resource Economics*. (ungated here)

7. **Advancing the estimation of future climate impacts within the United States**, with Corinne Hartin, Erin McDuffie, Karen Noiva, Marcus Sarofim, Jeremy Martinich, Sarah Barr, Jim Neumann, Jaqueline Willwerth, and Allen Fawcett. 2023. *Earth System Dynamics*.

8. **Comprehensive evidence implies a higher social cost of carbon**, with Kevin Rennert, Frank Errickson, Brian Prest, Lisa Rennels, Richard Newell, William Pizer, Cora Kingdon, Jordan Wingenroth, Roger Cooke, David Smith, Kevin Cromar, Delavane Diaz, Fran Moore, Ulrich Mller, Richard Plevin, Adrian Raftery, Hana Ševčíková, Hannah Sheets, James Stock, Tammy Tan, Mark Watson, Tony Wong, and David Anthoff. 2022. *Nature*.

9. **A market for snow: modeling winter recreation patterns under current and future climate**, with Peter Christensen. 2022. *Journal of Environmental Economics and Management*. (ungated here)

10. **Overlooked benefits of nutrient reductions in the Mississippi river basin**, with Amy Ando. 2020. *Land Economics*. (ungated here)

11. **Willingness-to-volunteer and stability of preferences between cities: Estimating the benefits of stormwater management**, with Amy Ando, Catalina Cadavid, and Noelwah Netusil. 2020. *Journal of Environmental Economics and Management*. (ungated here)

12. **Benefits of the fire mitigation ecosystem service in The Great Dismal Swamp National Wildlife Refuge**, with Emily Pindilli, and Dianna Hogan. 2017. *Journal of Environmental Management*. (ungated here)

BOOK AND
ENCYCLOPEDIA
CHAPTERS

1. **Customer Participation in Lead Pipe Replacement: A New Jersey Case Study**, with Heather Klemick, Ann Wolverton, Kristin Epstein, and Sarah Armstrong. 2026. *AWWA Editorial*. (ungated here)

2. **The social cost of greenhouse gases and its use in U.S. federal policy**, with Elizabeth Kopits, Wade Davis, Charles Griffiths, Alex Marten, Al McGartland, David Smith, and Elizabeth Spink. 2025. *Encyclopedia of Energy, Natural Resource, and Environmental Economics*. (ungated here)

FEDERAL,
REGULATORY,
AND OTHER
GOVERNMENT
PUBLICATIONS

The Social Cost of Greenhouse Gases

1. Estimates Incorporating Recent Scientific Advances. 2023.

2. Interim Estimates under Executive Order 13990. 2021.

Natural Capital Accounting

1. National Strategy to Develop Statistics for Environmental-Economic Decisions: A U.S. System of Natural Capital Accounting and Associated Environmental Statistics. 2023.

Federal Budget Chapters (Executive Office of the President)

1. Addressing the impacts of climate change: Adaptation planning and the climate benefits of federal investments. 2025.
2. Analysis of federal climate financial risk exposure. 2024.
3. Climate financial risk: The federal government's budget exposure to financial risk due to climate change. 2024.

Phasing Down Production and Consumption of Hydrofluorocarbons

1. 2024 and beyond Allowance Allocation Rule. 2023.
2. Protection of Stratospheric Ozone Rule. 2021.
3. Framework Rule. 2021.

PAPERS IN
PREPARATION

1. **Economic damages from climate change to U.S. populations: Integrating evidence from recent studies**, with Elizabeth Kopits, Daniel Kraynak, Lisa Rennels, David Smith, Elizabeth Spink, Charles Griffiths, Joseph Perla, Nshan Burns, and Michael Howerton (NCEE working paper here, and Arxiv working paper here)
2. **Robustness to model uncertainties drives more rapid CO₂ emissions reductions**, with Lisa Rennels, Frank Errickson, David Smith, Klaus Keller, and David Anthoff
3. **When it rains, it pours: severe weather events, flooding, and drinking water quality**, with Wes Austin, and Siyu Pan (working paper here)
4. **The circular economy of biosolids and its effect on drinking water quality**, with Patrick Walsh, and Wes Austin
5. **Valuing changes in aquatic biodiversity and recreation with a nationwide survey**, with Chris Moore, Stephen Newbold, Matthew Massey, and David Smith (working paper here)
6. **Contrasts and convergence of latent class and mixed logit models of discrete choice data**, with Chris Moore, Stephen Newbold, Matthew Massey, and David Smith
7. **A recreation demand model for mountain snowpack** (working paper here)
8. **Measuring the social cost of greenhouse gas emissions: Progress and a path forward**, with Elizabeth Kopits, Wade Davis, Charles Griffiths, Alex Marten, Al McGartland, David Smith, and Elizabeth Spink
9. **Wildfires degrade surface water quality and ecosystem services**, with David Smith and Michael Howerton
10. **Using discrete choice experiments as a tool for teaching consumer theory: A case study in an intermediate microeconomics course**

FEDERAL COMMITTEES, WORKGROUPS	Natural Capital Accounts and Environmental-Economic Statistics	2022- <i>present</i>
	Policy Working Group	
	Council of Account Leads	
	Environmental-Economic Accounting Classifications	
	American Innovation and Manufacturing Act Workgroups	2021- <i>present</i>
	Hydrofluorocarbons Phasedown, Technology Transitions, Allowance Allocation	
	Assessments of Federal Financial Climate Risk	2023-2025
	Executive Office of the President (OMB-CEES)	
	Interagency Working Group on the Social Cost of Greenhouse Gases	2021-2025
	Damage Functions, Discounting, Socioeconomic and Emissions Scenarios	
AWARDS	Ukraine Working Group on the Environment	2022-2024
	Ecosystem Services and Environmental Degradation	
	Federal Efforts to Address Harmful Algal Blooms and Hypoxia	2020-2022
	Government Accountability Office	
	Trudy A. Speciner Award for Advancing Environmental Protection	2023
	U.S. EPA, National Honors Award	
	Gold Medal for Exceptional Service	2023
	U.S. EPA, National Honors Team Award	
	Bronze Medal for Commendable Service	2023
	U.S. EPA, Office of Atmospheric Protection	
TEACHING EXPERIENCE	Cross-Cutting Group Award	2022
	Executive Office of the President of the United States, OMB	
	Rising Star Honors Award	2022
	U.S. EPA, Office of the Administrator	
	Most Outstanding Dissertation Award	2021
	University of Illinois, Department of ACE	
	Louis V. Logeman Teaching Award	2020
	University of Illinois, College of ACES	
	Environmental Economics	
	Georgetown University, DC (Econ 475)	FA22
	<i>Lecture:</i> 40 students	
	Undergraduate Credit: 3 Hours	
	Salisbury University, MD (ENVS 210)	SP21
	<i>Online:</i> 40 students	
	Undergraduate Credit: 3 Hours	
	Intermediate Microeconomics (ACE 398)	SP19, FA19
	University of Illinois at Urbana-Champaign, IL	
	<i>Lecture:</i> 65 students	
	Undergraduate Credit: 3 Hours	
	Teaching Evaluation Scores: 4.9/5	
	Microcomputer Applications (ACE 161)	SP18, SU18, FA18
	University of Illinois at Urbana-Champaign, IL	
	<i>Lecture:</i> 50 students, <i>Online:</i> 100 students	
	Undergraduate Credit: 3 Hours	
	Teaching Evaluation Scores: 4.8/5	
	Graduate Academy for College Teaching	SP19, FA19, SP20

University of Illinois at Urbana-Champaign, IL
Teaching Teachers how to Teach

REVIEWER	Proceedings of the National Academy of Sciences — Nature Climate Change — Nature Communications — Journal of Environmental Economics and Management — American Journal of Agricultural Economics — Land Economics — Environmental and Resource Economics — Resource and Energy Economics — Food Policy — Regional Science and Urban Economics — Landscape and Urban Planning — Land Use Policy — Ecological Economics — International Journal of Biometeorology — Environmental Development	
ADDITIONAL EXPERIENCE	Parthum Construction Ltd. Owner and Founder Tabernash, CO Design and build high-end residential remodels, manage large budgets (\$20k-\$1mil), supervise employees and all sub-contractors.	2005 - <i>present</i>
	Department of the Interior, U.S. Geological Survey Student Economist Reston, VA Supervisor: Emily Pindilli Inform adaptive management decisions within the U.S. interior and abroad through interdisciplinary and interagency collaboration.	2015 - 2020
REFERENCES	Available upon request.	
PROGRAMMING	•         	
LANGUAGES	• English (native), American Sign Language (conversational)	